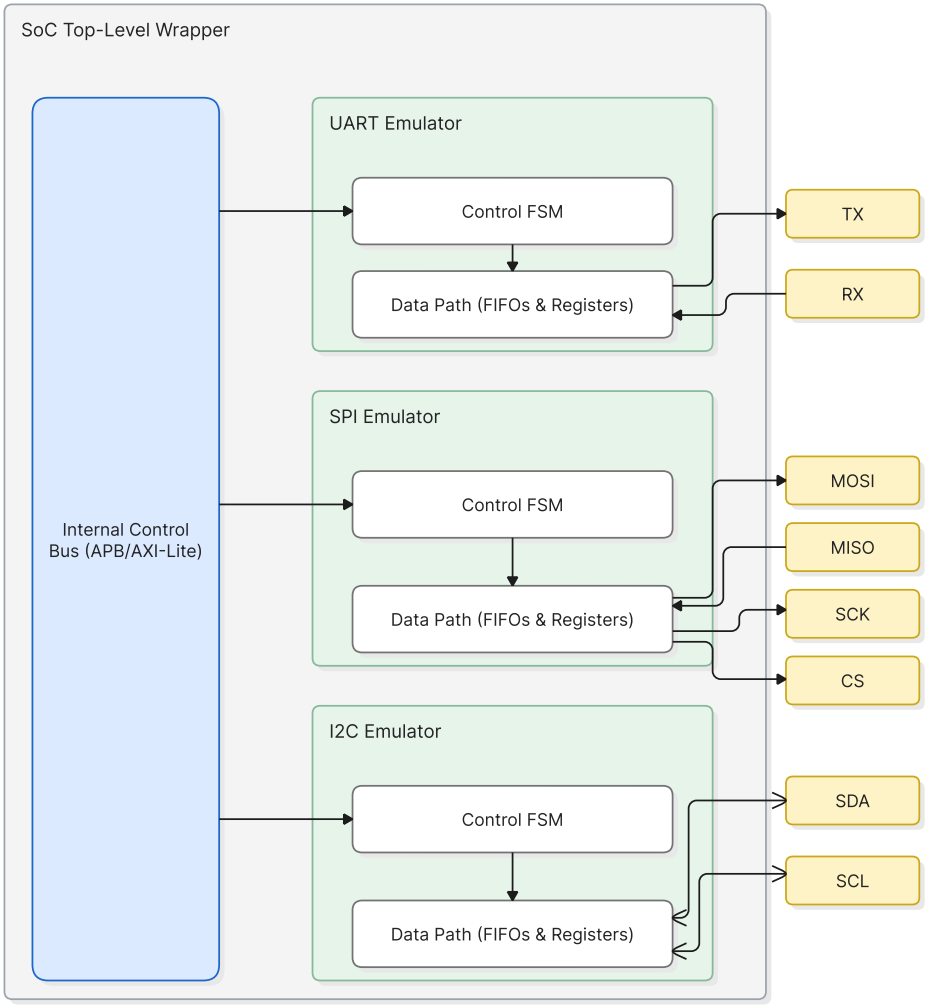
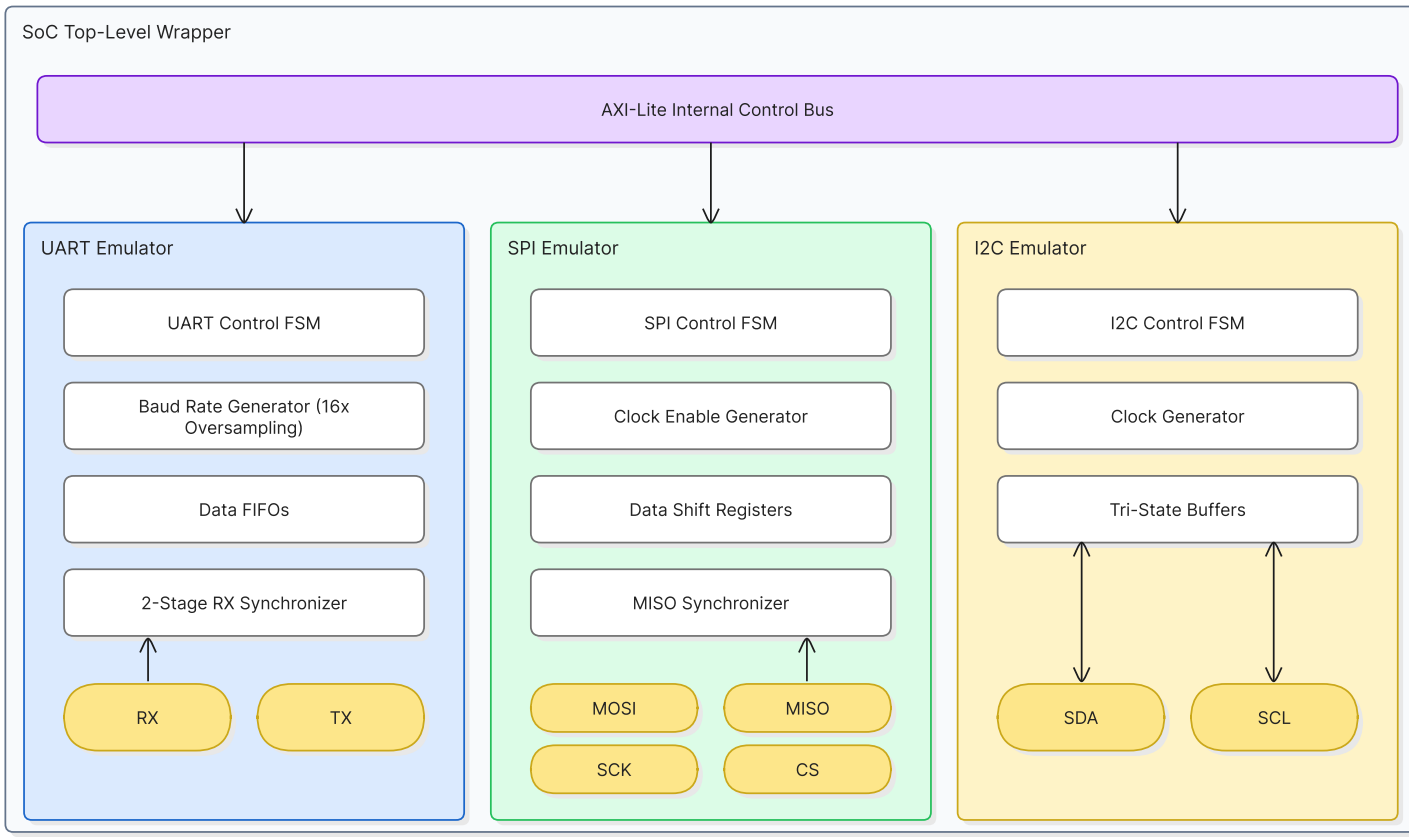

SoC Top-Level Wrapper
Internal Control Bus (APB/AXI-Lite)
UART Emulator
Control FSM
Data Path (FIFOs & Registers)
TX
RX
SPI Emulator
Control FSM
Data Path (FIFOs & Registers)
MOSI
MISO
SCK
CS
I2C Emulator
Control FSM
Data Path (FIFOs & Registers)
SDA
SCL


SoC Top-Level Wrapper
AXI-Lite Internal Control Bus
UART Emulator
UART Control FSM
Baud Rate Generator (16x Oversampling)
Data FIFOs
2-Stage RX Synchronizer
RX
TX
SPI Emulator
SPI Control FSM
Clock Enable Generator
Data Shift Registers
MISO Synchronizer
MOSI
MISO
SCK
CS
I2C Emulator
I2C Control FSM
Clock Generator
Tri-State Buffers
SDA
SCL